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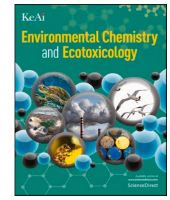

Research Paper

A systematic study on PM2.5 and PM10 concentration prediction in air pollution using machine learning and deep learning model

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ABSTRACT

High concentrations of PM10 and PM2.5 in the air pose significant risks to human health, making accurate forecasting crucial for air quality management and public safety. This study evaluates predictive models for particulate matter concentrations using air quality monitoring data from Maharashtra state (2019–2023). To determine their predictive effectiveness, various machine learning models, including Linear Regression, Decision Tree, Random Forest, and XGBoost, were compared against a deep learning-based Long Short-Term Memory (LSTM) model. To quantify their accuracy, the models were assessed using R^2 Score and Root Mean Square Error (RMSE). Our results demonstrate that LSTM significantly outperformed traditional machine learning models, achieving R^2 scores ranging from 0.99 to 0.998 across the five cities analyzed. In contrast, other models, including Random Forest, Decision Tree, and XGBoost, exhibited R^2 scores between 0.15 and 0.96, indicating lower predictive accuracy. The findings highlight that LSTM models can effectively capture complex temporal dependencies in air quality data, making them a more reliable approach for forecasting PM2.5 and PM10 concentrations. These results underscore the potential of deep learning in enhancing air quality prediction models and aiding environmental decision-making.

1. Introduction

The escalating impact of air pollution, intensified by fast-paced urban growth and industrial expansion, has made it one of the most pressing environmental challenges worldwide. Reliable forecasting of pollutant levels is vital to implement timely and effective strategies for mitigation and public health protection [60]. Extensive evidence gathered since the latter half of the 20th century highlights climate change as a persistent shift in weather patterns occurring at both global and regional scales [59]. Air pollution is a critical environmental challenge that affects numerous regions across the globe, posing significant threats to both human health and the natural ecosystem [1]. It poses a serious threat to human health and public safety [2]. Elevated concentrations of PM10 and PM2.5 in the air pose significant health risks, making accurate predicting particulate matter essential for mitigating their impact. As air pollution continues to deteriorate, accurately predicting PM2.5 concentration becomes a critical priority for safeguarding public health [3].

The increasing urbanization and industrialization across the globe, particularly in densely populated regions like Maharashtra, have escalated concerns about the detrimental effects of air pollution. Accurately forecasting air quality levels plays a pivotal role in safeguarding public health and guiding informed economic and policy decisions [4]. Urban air pollution is heavily influenced by a combination of factors such as industrial emissions, farming practices, expanding cities, transportation networks and ongoing infrastructure development [52,53]. Fine particulate matter, such as PM2.5 and PM10, has been identified as a critical environmental and public health issue due to its direct impact on human well-being, animal life, and vegetation. Air quality, which refers to the level of pollution-free air in a given area, is now considered a key indicator of sustainable development and urban health. Each year, a significant number of lives in India are lost due to the severe deterioration of urban air quality [5]. When present at concentrations above normal atmospheric levels, air pollutants can cause significant harm, including respiratory and cardiovascular diseases in humans, reduced

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visibility, and the deposition of harmful trace components in ecosystems. There is a limited body of research that specifically quantifies PM2.5 levels from wildfire smoke to assess their impact on hospitalizations due to respiratory and other illnesses, and existing studies are typically constrained by narrow geographic coverage and brief observational periods [56] [57] [58]. PM2.5 particles, in particular, are small enough to penetrate deep into the lungs, posing severe risks to public health. Wildfire smoke poses a greater threat to human health than typical air pollution due to its high concentration of hazardous substances, including fine soot particles and gases like carbon monoxide [55]. A major concern is the presence of PM2.5 tiny particles small enough to enter deep into the lungs and bloodstream, where they can trigger serious health issues [54]. While the harmful consequences of particulate matter are well-documented, there remains a pressing need for innovative approaches to predict and manage air quality effectively [6]. This study addresses this gap by focusing on developing predictive models that can enable timely interventions and support air quality management strategies [7]. By advancing techniques to accurately forecast PM concentrations, the research aims to contribute to sustainable urban living and mitigate the adverse effects of air pollution. A link between adverse health consequences and increased PM10 or PM2.5 atmospheric concentrations has been found in several research [8].

Due to its extremely small aerodynamic diameter ($\leq$2.5 μm for PM2.5 and $\leq$ 10 μm for PM10), fine particulate matter can penetrate deeply into the respiratory system, reaching the bronchioles and alveoli, where it can interfere with lung gas exchange and contribute to respiratory dysfunction [9]. Air quality forecasting is essential to help individuals make decisions that limit their exposure to PM2.5, such as whether to engage in outdoor or indoor activities [10].

Taiwan's Environmental Protection Board has set up over 65–75 air quality sensors to evaluate air quality on daily bases and educate the public. How to accurately predict air pollution is a topic that is getting more and more crucial [11]. In many recent research, air quality forecasts have been used to signal air pollution. The historical statistics on manufacturing chimney emissions, steam locomotive emissions, and global pollution sources significantly correspond with the current PM 2.5 concentration. Forecasting PM2.5 levels is challenging due to the complexity of deriving accurate predictions from historical ground-level air quality and weather data, highlighting the need for advanced predictive models and techniques.

The public and governmental organizations can use air pollution forecasting to help them prepare for extreme pollution episodes and make decisions. Accurate and trustworthy estimates of ambient PM10 and PM2.5 concentrations are necessary to improve air quality and safeguard public health [12]. Various approaches have been employed to predict AQI, including statistical, deterministic, physical, and machine learning (ML) models. Traditional approaches focused on statistics and probability are incredibly time-consuming and ineffective. It has been demonstrated that the ML-based AQI prediction models are more dependable and consistent. The collecting of data has been made simpler and more accurate thanks to modern technology and sensors.

Machine learning and deep learning algorithms are able to conduct the meticulous analysis necessary for precise and trustworthy predictions based on such a massive amount of environmental data. In-depth analysis is required to assess raw data and identify pollutants. These algorithms provide accurate future PM value projections that allow for the application of the necessary safety precautions. There are three different learning algorithms for machine learning, a subset of artificial intelligence: supervised, unsupervised, and reinforcement learning. Linear Regression, Random Forest, XG Boost, Decision Tree, Nearest Neighbour, SVM, kernel SVM, Naive Bayes, and other algorithms fall within the category of supervised learning algorithms [13]. XG Boost and Random Forest produce better results in comparison to all other methods. However, employing an ML model will result in high RMSE (error); hence, utilizing a deep learning model is necessary to provide better results with lower errors. The nonlinear, cyclical, seasonal, and sequential correlations between pollutant data can be solved using the deep learning architecture and challenges with air pollution prediction [14].

LSTM is designed to more successfully learn and maintain long-term dependencies from time-series data than shallow artificial neural network architecture [15]. Measurements of pollution levels and other sequence-dependent variables are great candidates for the internal memory of the LSTM [30]. Selecting the hyperparameters for the LSTM model is a crucial step in building a DL model. The model's processing, overfitting, and training processes are influenced by the selection of these hyperparameters, which has an impact on the model's accuracy.

It can be time-consuming and error-prone to select the ideal total layer numbers, per layer number of cells, units, size of group, type of activation, and related parameters [16]. In order to build a more perfect model for forecasting air pollution with the help of the Meta-heuristic's technique, the issue of choosing hyper-parameters, specifically the window size and total LSTM unit numbers, is addressed [17]. Further in this paper, the related works section goes on to address numerous case studies by various authors in this field, followed by a discussion of the dataset and its pre-processing, and a discussion of the various models we used to analyze the dataset for this paper. Further, we discuss challenges and future work in this paper and our domain. Next, we have mentioned all results and visualization of models that we had applied to the dataset; last, we have a conclusion.

2. Related works

Liang et al., 2024 proposed research on air pollution prediction using LSTM networks, which effectively capture long-term dependencies in sequential data. A key challenge lies in optimizing LSTM architectures and hyperparameters, typically determined manually. To address this, the study proposes HGA-LSTM, a novel method combining Genetic Algorithm (GA) with adaptive local search for simultaneous optimization of LSTM architecture and hyperparameters [19]. Based on findings from epidemiological cohort studies, fine particulate matter (PM2.5) particles with diameters smaller than 2.5 μm can remain airborne for extended periods and act as carriers for harmful chemicals and microbial agents [56]. A specialized crossover mechanism is introduced to handle variable-length LSTM representations. Experiments on UCI datasets demonstrate that HGA-LSTM outperforms widely-used LSTM-based and non-LSTM methods, achieving lower mean square/absolute errors. Additionally, HGA-LSTM surpasses standard GA in convergence and accuracy, showcasing improved search capability through the hybrid approach [18].

Tsai et al., 2018 addressed the critical issue of PM2.5 air pollution, which is closely linked to human health. With advancements in technology, accurate forecasting of PM2.5 has become increasingly important. The researchers proposed a forecasting approach using a Recurrent Neural Network (RNN) with Long Short-Term Memory (LSTM) to tackle this. The model was developed using Keras, a high-level neural network API running on TensorFlow. Training data from Taiwan's Environmental Protection Administration (EPA) for 2012–2016 comprised 20-dimensional inputs, with testing data from 2017. Experiments were conducted to predict PM2.5 concentrations for the next four hours across 66 monitoring stations in Taiwan. The results demonstrate that the proposed RNN-LSTM approach effectively forecasts PM2.5 values, providing a valuable tool for managing and mitigating air pollution impacts. This methodology highlights the potential of deep learning techniques in environmental monitoring. The MAE might vary between 1.644 and 5.543 within an hour of forecasting. Even the long-term prediction correctly foresees a rise in PM2.5 levels [20].

Barthwal and Goel, 2024 did a study addresses the critical issue of urban air quality degradation in India, which claims numerous lives annually, by proposing a deep learning framework to analyze and predict air conditions in Delhi. Utilizing meteorological and air quality data, including carbon monoxide (CO) and particulate matter (PM2.5

and PM10) levels, from monitoring stations across 14 locations, an extensive 1765-day Air Quality Index (AQI) dataset (2014–2020) was created and split into training (80 %) and testing (20 %) sets. The AQI values, categorized into five pollution severity levels, were used to train and evaluate deep learning models. The study developed and compared deep convolutional neural networks (DCNN) and a hybrid DCNN-long short-term memory (DCNN-LSTM) architecture, leveraging DCNN's ability to extract key features from time-series data and LSTM's proficiency in capturing temporal correlations. The DCNN-LSTM model, integrating the strengths of both architectures, achieved superior predictive performance with an accuracy of 97.48 %, an F1 score of 97.48 %, and an AUC of 0.97, outperforming the standalone DCNN model (accuracy: 94.48 %). These findings highlight the potential of deep learning in improving air quality forecasting, offering a robust tool to address air pollution challenges in urban environments.

Kushwah and Agrawal, 2024 address a critical issue of air quality prediction in urban areas, which has significant implications for public health and economic decisions. It proposes a novel hybrid approach, EMD-LSTM-Bayesian, combining empirical mode decomposition (EMD), long short-term memory networks (LSTM), and optimization techniques such as random search and Bayesian optimization [21]. EMD is employed to decompose the data into subseries for reduced complexity, while LSTM predicts time series with optimized hyperparameters. The proposed model achieves superior performance, with the lowest MAE (0.385) and RMSE (0.533), compared to LSTM alone, which recorded the highest MAE (0.593) and RMSE (0.804). Experimental results demonstrate that EMD-LSTM-Bayesian improves accuracy by 35.07 %, 20.94 %, 4.22 %, and 27.35 % over LSTM, LSTM-Random Search, LSTM-Bayesian, and EMD-LSTM, respectively, on the 2013 dataset. The findings underscore the model's efficacy in enhancing air quality predictions, paving the way for better-informed policy-making and public health interventions [22].

Li et al., 2022 did research that addressed the challenge of air pollution by proposing a novel deep learning-based model, CBAM-CNN-Bi LSTM, for accurate PM2.5 concentration prediction. The methodology integrates the convolutional block attention module (CBAM) for extracting feature relationships and spatial distribution characteristics, a convolutional neural network (CNN) for feature learning, and a bidirectional long short-term memory (Bi LSTM) network for capturing temporal variations and addressing long-term dependencies. The model was evaluated against other approaches using real datasets, demonstrating superior performance in predicting PM2.5 concentrations across multiple timeframes. The results show that the model excels in forecasting tasks from 1 to 12 h and achieves reliable predictions for 13 to 48-h tasks, offering a robust solution for air quality management and pollution warning systems [23].

Zhou et al., 2014 have proposed that Since PM2.5 contains tiny solid or liquid particles that can be breathed deeply into a person's lungs, exposure to high levels of PM2.5 can cause major health problems. Regulatory plans must consider daily estimates of PM2.5 concentrations to inform the public and restrict social activities before dangerous events. Based on data pre-processing and analysis, this study proposes the first hybrid EEMD-GRNN model for one-day-ahead PM2.5 concentration prediction. The EEMD component divides the original PM2.5 data into several intrinsic mode functions (IMFs), whereas the GRNN component predicts each IMF. In Xi-an, China, the model is tested using data from 2013 to 11-02 to 2013-11-21 and trained using data from 2013 to 01-01 to 2013-11-01. Rapid air quality warning systems can be built using the hybrid model's accurate and timely results. The following findings are reached after looking at model performance metrics: (1) The EEMD-GRNN model's MAE, MAPE, RMSE, and IA are 19.89, 28.15, 29.45, and 0.8442, respectively. (2) Despite having a comparatively less predictive precision, the PCR model has a good prediction accuracy [24].

Hooyberghs et al., 2005 had stated that the health effects of ambient particulate matter (PM) have drawn the attention of environmental scientists in recent years, and much research is being done to better understand this phenomenon and predict ambient PM concentrations [25]. The building of a neural network system is described in this article to predict the typical daily PM 10 concentrations in Belgium one day in advance. This study is based on data acquired from ten monitoring sites between 1997 and 2001 as well as climatic variable models from the ECMWF. The height of the border layer was the greatest valuable input variable. Currently, the daily average threshold of 100 g/m^3 is tracked using an online model based on this parameter. We were only able to somewhat enhance the model's performance by introducing new input parameters. We are dealing with what is known as a regression problem. A bunch of given input variables must be used to create an unknown variable that, on average, resembles the requisite PM10 day1 concentration. A neural network must be built for each evaluating site to match a function between the chosen inputs and the PM10 day1 target [26]. The input vector and its matching target are constructed from previous data in a set of records. A portion of the data used for non-linear curve fitting is subsequently required to build the NN. The remaining data in the dataset often acts as a separate test set against which the effectiveness of the neural network is verified. As a result, we conclude that changes in anthropogenic sources have a minor impact on daily variations in PM 10 concentrations in metropolitan regions of Belgium.

Chaloulakou et al., 2003 proposed a study that explored the potential of neural networks as predictive tools for daily average particulate matter (PM10) concentrations, addressing the critical issue of urban air pollution and its impact on human health. Utilizing a comprehensive two-year dataset from a fixed monitoring site in Athens, Greece, and employing meteorological variables as key inputs, the study developed and evaluated both neural network and traditional multiple linear regression models. Comparative analysis revealed that neural networks demonstrated superior performance, with a reduction in prediction error (root mean square error) by 8.2–9.4 % and a decrease in false alarm rates by 7–13 % during episodic prediction scenarios. These results suggest that, when appropriately trained and structured, artificial neural networks (ANNs) can effectively meet the demands of accurate particulate pollution forecasting. By outperforming regression models in both predictive accuracy and episodic reliability, ANNs show great promise for enhancing public health awareness and improving air quality management strategies in urban areas [27].

Wu et al., 2023 researched the pressing issue of severe air pollution, which poses a significant threat to public health and safety, by developing an advanced ISSA–LSTM model for accurate air quality index (AQI) prediction. The proposed methodology integrates three key components to enhance prediction accuracy and efficiency: random forest (RF) and mRMR for selecting the most influential variables affecting AQI, the improved sparrow search algorithm (ISSA) for optimizing the hyperparameters of the long short-term memory (LSTM) network, and the LSTM itself for performing AQI predictions. The model eliminates irrelevant variables by leveraging RF–mRMR [2], improving both speed and accuracy. ISSA fine-tunes the LSTM hyperparameters, significantly enhancing the network's performance. Extensive comparative experiments validate the ISSA–LSTM model's effectiveness, demonstrating superior performance with reduced RMSE and increased R^2 values compared to traditional machine learning approaches. Furthermore, the model achieves minimal prediction errors across varying time steps, highlighting its robustness and adaptability. These results establish the ISSA–LSTM model as a reliable and practical tool for AQI forecasting, which is critical for implementing pollution control strategies and guiding residents in making informed activity choices.

Oprea et al., 2017 stated that Artificial intelligence-based forecasting techniques are more accurate than deterministic techniques in concurrent air pollution predicting systems. Our research aims to identify the best artificial intelligence model for predicting real-time PM2.5 air pollution in metropolitan regions. ANNs and ANFIS have been found to be the most effective computational intelligence models for predicting particulate matter (fraction PM2.5) air pollution. Based on the results of the experiments, we thoroughly analyze the identification of the optimum ANN architecture. AirBase database records with hourly PM2.5 concentration readings were used for the experiments [29]. The statistical measurements that have been calculated include the correlation coefficient, agreement index, mae, and rmse. After carefully comparing the ANN and ANFIS models, both of which showed promise in real-time forecasting systems, the artificial intelligence-based PM2.5 forecasting model was determined to be the best choice. The standard protection limit levels for human health are detectable by the ANN PM2.5 forecasting model [28].

Dai et al., 2021 proposed a study that addresses the increasing severity of air pollution in China by proposing an advanced model for predicting PM2.5 concentration, leveraging spatio-temporal features for enhanced accuracy. The methodology integrates XGBoost for feature extraction, MSCNN for capturing local spatio-temporal relations, and LSTM optimized via a genetic algorithm to model long-term dependencies. Using a dataset of hourly pollutant concentrations and meteorological data from Fen-Wei Plain in 2020, the proposed XGBoost-MSCGL model is compared against baseline models like MLP, CNN, LSTM, and their combinations. Results indicate significant improvements, with RMSE reduced by 39.07 %, MAE by 42.18 %, and R^2 increased by 23.7 % over single models, while also outperforming feature-selected models with an average RMSE reduction of 15 %. The findings demonstrate the model's superior accuracy, generalization ability, and comprehensive prediction of PM2.5 concentration, providing a valuable tool for air quality management.

Jlifi et al., 2024 researched sustainable urban development by addressing traffic congestion, a significant contributor to environmental degradation and climate change. Traffic forecasting, which leverages historical and real-time data to predict future patterns, is key to enhancing traffic flow management. The study proposes an innovative approach using air quality data to predict traffic congestion rates. A novel ensemble voting model was developed, combining Long Short-Term Memory (LSTM) and Polynomial Regression (PolReg) models with a newly designed threshold-based voting algorithm. Hyper-parameters were fine-tuned using Genetic Algorithms to address the non-stationarity of time series data. The proposed framework was benchmarked against existing methods, demonstrating superior performance with learning curves validated by Mean Absolute Error of 0.04, R-Squared of 0.93, and Root Mean Square Error (RMSE) of 0.05. These results confirm the model's efficacy, providing a robust tool for smarter, environmentally conscious traffic management solutions in sustainable cities [31].

Joy et al., 2022 proposed a study that focused on addressing the severe air pollution problem in Manila, where environmental issues linked to poor air quality pose significant health risks, particularly to vulnerable groups. To mitigate these impacts, the researchers developed the first air quality forecasting model in the Philippines using an Artificial Neural Network (ANN). The model leverages a feed-forward neural network architecture to predict PM2.5 and PM10 pollutant concentrations, using data collected from Manila's Mehan Garden monitoring station. The study emphasizes the importance of forecasting air quality indicators to inform the public about whether the air is safe or hazardous. The ANN model was evaluated using performance metrics such as Mean Squared Error (MSE), Mean Absolute Percentage Error (MAPE), and Mean Absolute Error (MAE). The results showed high predictive accuracy, with R^2 values close to 1, demonstrating the model's reliability in accurately forecasting pollutant levels. By providing real-time and accurate predictions, this model represents a vital tool for protecting public health and supporting policymakers in addressing the challenges posed by air pollution in urban environments. The study underscores the potential of advanced machine learning techniques to improve air quality management and promote healthier living conditions in cities like Manila [32].

Fernando et al., 2011 had introduced that Urban airshed planning and regulatory administration frequently use deterministic photochemical air quality models. Routine air quality forecasting is extremely expensive due to these systems' complexity and computer-intensiveness. Stochastic approaches, which hand off decision-taking to AI based on neural networks comprised of synthetic "nodes" able to " learn by training" using historical data, are on the rise. In order to forecast the particle amount at a daily monitoring site in Arizona, this study deployed a neural network. In conclusion, neural networks can be implemented much more quickly, cheaply, and easily without sacrificing accuracy. A network of automated monitoring stations can be utilised with neural networks to provide quick alarm systems for air quality.

Balaraman et al., 2022 had introduced that air pollution poses a serious hazard to human life in India. 77 % of Indians were exposed to Particulate Matter (PM2.5), which caused 6,700,000 deaths nationwide in 2017. This study employs a powerful deep learning technique called the Long Short-Term Memory (LSTM) model to forecast PM2.5. The predictive model's efficacy is evaluated using the R2-score and RMSE. The models are run using continuous ambient air quality data collected at 15-min intervals between May 1 and April 30 from 14 Indian sites. The best outcomes come from models with 64 epochs of adjustable parameters and a 32-person group size. There were accurate forecasts of PM2.5 concentration from all three LSTM model versions. According to the experimental results, the R2 value remains at 0.90 for all LSTM model iterations. The model's strong R2 values and low RMSE values demonstrated its dependability. The suggested strategy thus alerts the general people to the severity of air pollution in India and implores them to adopt safety measures. Pollution levels can be assessed by urban planners for planning and decision-making purposes.

In this paper, Voukantsis et al., 2011 stated a methodology based on some computational intelligence approaches, specifically principal component analysis and ANN, to compare meteorological data on air quality and projected concentrations of pertinent environmental variables (air pollutants). These techniques are used to examine data gathered from Helsinki, Finland, and Thessaloniki, Greece. Principal component analysis was used to assess how the two chosen cities' air pollution patterns differed from one another. After that, we created air quality forecasting models for both research regions. In order to choose the input variables for forecasting models, we developed and deployed a novel hybrid approach that blends linear-regression with ANN. The kappa index for predicting daily average PM 10 concentrations for both locations neared 60 %, the agreement index between calculated and simulated daily average PM-10 amount ranged from 0.80 to 0.85. In comparison to earlier studies, these statistical parameters show an improvement in parameter predictions for air quality. Despite the significant differences between the two urban settings under examination, the models' ability to predict the daily mean PM 10 concentrations in the two cities showed no discernible variance. For the purpose of creating and comparing profiles of urban air quality, we suggest a data-driven approach. In two European cities—Helsinki and Thessaloniki—this methodology is assessed. Despite the disparities between the two areas under study, the neural network modeling approaches delivered reliable air quality predictions. The prediction models were improved using a brand-new hybrid input selection technique [33].

Biancofiore et al., 2017 proposed that one of the contaminants in the air that has the potential to significantly affect human health is particulate matter (PM). Using recursive and non-recursive neural network models, a multi-linear regression model, and a model of neural networks with and without recursive architecture are used to analyze data gathered over three years in an urban region on the Adriatic coast. The average daily PM10 concentration is predicted one to three days in advance using meteorological variables and measurements of the concentration of PM10. The usefulness of PM forecasts as an operational tool is evaluated by comparing the findings against health protection air quality limits [34]. Improved projections have been studied when carbon monoxide amount is incorporated as a model input component. Meteorological information, PM10 concentration, and CO concentration are used as inputs in all models to forecast the PM2.5 concentration in order to simulate a shortage of PM2.5. The neural network can forecast PM2.5 concentrations even without using PM2.5 as one of the input parameters, as shown by the comparison of the observed and predicted PM2.5 concentrations [35].

Kök et al., 2018 stated that in recent years, the Internet of Things (IoT) has emerged as an interesting research issue in a variety of industries, including business, education, and industry. IoT services and applications in smart cities encourage environmentally friendly urban living. IoT uses information and communication technology to improve the consciousness, interaction, and efficiency of smart cities. Along with the growth of IoT-based smart city applications, these apps are producing more and more data. In order to encourage sustainable growth, governments and city stakeholders take proactive measures to assess these data and anticipate future implications. Deep learning approaches have already been used to resolve a number of massive data forecasting challenges in the area of prediction. This motivates us to use deep learning algorithms to forecast IoT data. This study provides a brand-new deep learning model to analyze data from Internet of Things (IoT) smart cities [36]. In order to estimate future air quality values in a smart city, we propose a novel model based on networks with long short-term memory (LSTM). The evaluation's findings indicate that the suggested approach can address more prediction issues in smart cities.

Srivastava and Kumar, 2023 gave research that addresses the growing global concern of air pollution and proposes an AI-based prediction model to tackle this challenge. The study introduces the novel XRSTH-LSTM model, fine-tuned for pre-processing, attribute extraction, and air-quality index prediction to enhance accuracy and reduce computational cost. The model was trained on India's Air Quality Data (2015–2020) from Kaggle, which includes hourly and daily AQI readings across multiple cities. Performance metrics such as MSE, MAPE, RMSE, precision, recall, and F-measure were used to evaluate the model, with additional robustness testing via negative predicted value and Matthew's correlation coefficient. The model achieved an accuracy of 98.52 % and a precision of 99.79 %, outperforming state-of-the-art approaches by 0.74 %. The results demonstrate the proposed method's superior efficiency and accuracy in predicting air pollution, marking a significant advancement in AI-driven environmental solutions [37].

Gokhale and Raokhande, 2008 stated that a significant factor in the growing air pollution issues of modern society is urban transport and how it interacts with the metropolis. While CALINE3 has the lowest absolute bias (AB) and fractional bias (FB) during the winter season, CAL3QHC has the lowest normalized mean square error (NMSE) for both particle sizes. M- CALINE3 and CAL3QHC have the greatest d values for PM2.5, whereas GFLSM has the highest d values for PM10. In Guwahati, the city in northeastern India with the highest population growth, the study showed the performance assessments of three models for projecting vehicle exhaust emissions at the busiest urban crossroads. According to the overall assessment study, CAL3QHC outperformed CALINE3 for PM2.5, whereas CALINE3 outperformed both for PM10 and PM2.5 [38]. The relationships between observed and simulated PM10 and PM2.5 values for the CALINE3 and CAL3QHC models were good. The CALINE3 and CAL3QHC models regularly beat the M-GFLSM model for PM10, with the exception of low wind speeds (b1 m s1), where the M-GFLSM model tended to perform better for PM10 and may be improved by integrating more dependable emission factors for both "idle" and "free-flowing" circumstances. The fact that the models used in this inquiry only approximated the influence of vehicle sources may be the cause of the inconsistency between the observed and predicted results [39].

3. Models employed for this paper

3.1. Decision tree

Decision Tree is a non-parametric, supervised learning algorithm widely used for classification and regression tasks. It works by splitting the dataset into subsets based on the feature that provides the maximum information gain or minimum impurity. The tree is built in a top-down manner, with each node representing a decision based on a feature, and each leaf node providing a prediction or output. Decision Tree employs pruning methods to prevent overfitting, ensuring the model remains generalizable to unseen data. It simplifies decision-making by learning interpretable rules directly from the data.

Various algorithms, such as CART (Classification and Regression Trees), ID3 (Iterative Dichotomiser 3), C4.5, and MARS (Multivariate Adaptive Regression Splines), are used to optimize Decision Tree construction, with each tailored for specific data types and requirements. For example, CART constructs binary trees and uses Gini Impurity or Mean Squared Error for node splits, while ID3 and C4.5 utilize entropy and information gain.

In the context of time-series forecasting, Decision Tree Regressor is particularly advantageous because it does not rely on assumptions about the underlying data distribution. It handles non-linear relationships effectively and provides reasonable results without extensive hyper-parameter tuning. The Decision Tree predicts the target variable by following a path from the root to a leaf node, guided by splits that minimize impurity or maximize homogeneity.

The key formula used in Decision Tree algorithms is the **entropy** or impurity measure, which quantifies the disorder or uncertainty in the data at each node. For a dataset SSS with nnn classes, the entropy is calculated as:

$$H(S) = \sum_{i=1}^{c} -p_i \, log_2(p_i) \tag{1}$$

where pi is the proportion of instances belonging to class i.

Decision Tree Regressors in time-series forecasting are highly flexible, capable of capturing complex temporal patterns. Despite their standalone effectiveness, they are often used as a base model in ensemble methods such as Random Forest and Gradient Boosting, where multiple decision trees combine to improve robustness and accuracy [40]. The simplicity, interpretability, and versatility of Decision Trees make them an essential tool in predictive modeling and forecasting tasks.

3.2. XGBoost

Gradient Boosting Regressor is an ensemble learning method that builds multiple weak models (typically decision trees) and combines them iteratively to improve performance and create a strong predictive model. **Extreme Gradient Boosting (XGBoost)** is an advanced and optimized version of Gradient Boosting, designed for both speed and accuracy. It incorporates system optimization techniques and algorithmic improvements, making it faster and more efficient than traditional Gradient Boosting (GB). XGBoost is particularly effective in regressive predictive models, often outperforming other ensemble methods due to its ability to handle sparse data, prevent overfitting through regularization, and leverage advanced techniques like tree pruning and parallel computation.

In the context of predicting the Air Quality Index (AQI), XGBoost excels because it uses a **similarity score** to evaluate and prune decision tree nodes dynamically. This ensures efficient predictions by focusing only on the most significant splits, thereby reducing complexity while maintaining accuracy. Additionally, XGBoost employs a **Gradient Boosting Regressor**, which uses the Mean Squared Error (MSE) as a cost function to evaluate the model's performance. It minimizes errors by iteratively refining predictions based on residuals from previous iterations.

A key feature of XGBoost is its use of a **learning rate (vvv)** to control the contribution of each weak learner to the overall model. In time-series data, where bias errors may arise in the training data, XGBoost helps to nullify such errors through its iterative process. At each step, an initial prediction F0 is adjusted by adding a scaled prediction from the weak learner (v·y1) to refine the final predictions. The formula can be expressed

$$F = F0 + v \cdot y1 \tag{2}$$

Here, F0 represents the initial prediction, v is the learning rate, and y1 is the contribution of the first weak learner. This process is repeated for each iteration, with subsequent learners correcting the residuals from the previous learners.

XGBoost also incorporates regularization parameters (λ,γ) to penalize the complexity of the model, further reducing the risk of overfitting. The optimization objective in XGBoost combines both the loss function (e.g., MSE) and a regularization term to balance the trade-off between model complexity and accuracy.

Overall, XGBoost stands out as a powerful and efficient algorithm for predictive tasks, particularly in handling large datasets, time-series predictions, and tasks requiring high accuracy. Its combination of gradient boosting, regularization, and advanced optimization techniques makes it a top choice for machine learning practitioners.

3.3. Random Forest Regressor

Random Forest is an ensemble learning method that builds multiple decision trees during training and aggregates their outputs to produce a final, more accurate prediction. Each tree in the Random Forest is trained on a randomly selected subset of the data using a technique called bootstrapping, where samples are drawn with replacement. This approach ensures diversity among the trees, making the model robust enough to be overfit. The results from individual trees are then averaged (for regression tasks) or taken as a majority vote (for classification tasks) to improve predictive accuracy. In predicting air quality, the Random Forest-based approach, RAQ (Random Forest for Air Quality), implements bootstrapping and bagging methods to reduce variance and minimize errors, thereby generating consistent and reliable results.

To ensure randomness, tree-growing and splitting criteria, such as the selection of features and thresholds, are randomized, which helps avoid overfitting on any feature set. he algorithm also uses entropy as mentioned in the equation below to evaluate the splits at each node, just like in a Decision Tree, ensuring that the data is divided effectively to maximize the predictive capability. Random Forest also considers a subset of features at each split, which helps handle high-dimensional datasets and prevents trees from becoming too similar. This ensemble method is particularly effective for air quality prediction because of its ability to model complex interactions between pollutants and other variables while also being robust to noise and outliers in the data. By aggregating multiple trees, Random Forest provides a balance between bias and variance, making it a powerful tool for accurate and reliable air quality forecasting.

$$E(s) = \sum_{i=1}^{c} -p_i \log_2(p_i) \tag{3}$$

$$p(c_i) = \frac{N_i}{\sum_{i=1}^{k} N_i} \tag{4}$$

Where p(ci) represents probability of the class and *Ni* represents total number of samples.

3.4. Long Short Term Memory

Long Short-Term Memory (LSTM) is a specialized type of Recurrent Neural Network (RNN) designed to overcome the limitations of traditional RNNs, particularly their inability to handle long-term dependencies in sequential data. Unlike standard RNNs, which struggle with the vanishing gradient problem, LSTMs incorporate a memory cell and gating mechanisms that allow them to retain or forget information over extended periods. This makes them highly effective for forecasting and predictive analysis, where past information significantly impacts future predictions.

An LSTM model processes input in sequential data or lists of features. Air quality forecasting, for example, can anticipate inputs like PM2.5 and PM10 levels, temperature, humidity, wind speed, concentrations of gases, and other environmental factors. LSTMs capture complex temporal patterns and relationships among variables by analyzing these sequential features, enabling precise and reliable predictions. Each LSTM unit consists of three gates—input, forget, and output gates—that dynamically control the flow of information, ensuring the model focuses on relevant features while discarding irrelevant data.

In the context of air quality prediction, the LSTM model learns the intricate interactions between pollutants and meteorological variables, leveraging its neurons to make accurate forecasts. The model adjusts its parameters based on the training data, gradually refining its predictions. With its ability to capture both short-term fluctuations and long-term trends, the LSTM excels in generating strong and consistent results, making it a powerful tool for applications where understanding temporal dependencies is critical.

$$i_t = \sigma(w_i[a_{t-1}, x_t] + b_i) \tag{5}$$

$$j_t = \sigma(w_j[a_{t-1}, x_t] + b_j) \tag{6}$$

$$k_t = \sigma(w_k[a_{t-1}, x_t] + b_k) \tag{7}$$

Where 'i_t' represents input gate.
'j_t' represents forget gate.
'k_t' represents output gate.
'σ' represents sigmoid function (activation function).
'w_x' represents weight for the representative gate(x) neurons.
'a_{t-1}' represents output of the previous lstm block.
'x_t' represents input at current timestamp.
'b_x' represents biases for the respective gates(x)

4. Methodology

4.1. Dataset

4.1.1. Data collection

This study utilizes air quality data obtained from the Central Pollution Control Board (CPCB), a statutory organization established in September 1974 under the Water (Prevention and Control of Pollution) Act, 1974, and later granted additional powers through the Air (Prevention and Control of Pollution) Act, 1981. As India's principal environmental regulatory body, CPCB monitors air and water pollution, enforces environmental standards, and advises the central government on pollution-related matters (Source: https://airquality.cpcb.gov.in/ccr/#/caaqm-dashboard-all/caaqm-landing/caaqm-data-availability) [61].

Air quality monitoring is a critical component of environmental management, allowing policymakers and researchers to track pollution trends, identify sources, and assess the effectiveness of air quality regulations. To support these objectives, CPCB launched the National Air Monitoring Programme (NAMP) to evaluate current air quality levels, detect pollution hotspots, and regulate emissions to ensure compliance with established air quality standards. Additionally, NAMP serves as a vital source of background data for industrial site selection, urban planning, and environmental impact assessments, helping to guide policy decisions and mitigate pollution-related health risks.

For this study, we use real-time air quality data from CPCB's official database, focusing on key air pollutants, including PM10 and PM2.5, recorded daily across multiple monitoring stations. A representative day for air quality assessment requires at least sixty (60) minutes of valid data collection to be considered reliable. The dataset spans from January 2019 to March 2023, covering a period of over four years, ensuring a robust dataset for machine learning-based predictive modeling (Source: https://airquality.cpcb.gov.in/ccr/#/caaqm-dashboard-all/caaqm-landing/caaqm-data-availability).

Given the specialized nature of the Machine Learning (ML) modeling process, the dataset was structured into two subsets:

1. Training Set – Used for model development, parameter tuning, and learning patterns from historical air quality trends.
2. Testing Set – Used to validate the trained model's performance and assess its predictive accuracy on unseen data.

The complete dataset, incorporating both training and testing sets, was used to develop and evaluate the model's ability to predict future air quality conditions.

For this study, we focus on air quality measurements from five major Indian cities:

- Aurangabad
- Mumbai
- Nashik
- Thane
- Pune

The dataset is a time-series collection, comprising information on various pollutants, meteorological parameters, and temporal attributes. The key pollutants included in the dataset are:

- Particulate Matter (PM10, PM2.5)
- Carbon Monoxide (CO)
- Nitrogen Dioxide (NO_2)
- Nitrous Oxide (NO)
- Ammonia (NH_3)
- Sulfur Dioxide (SO_2)
- Ozone (O_3)
- Benzene (C_6H_6)
- Toluene ($C_6H_5CH_3$)
- Ethylbenzene ($C_6H_5C_2H_5$)
- Xylene ($C_6H_4(CH_3)_2$)
- m,p-Xylene

Along with pollutant concentration levels, the dataset also contains supplementary information crucial for time-series analysis, including:

- City Name
- Date (DD/MM/YYYY format)
- Day of the Week
- Month and Year
- Relative Humidity
- Wind Speed
- Wind direction
- Temperature of the city
- Barometric Pressure
- Vertical Wind Speed
- Solar Radiation

This structured dataset serves as the foundation for our study, enabling the development of predictive models to assess and forecast air pollution trends. By leveraging machine learning algorithms and deep learning architectures, we aim to provide accurate air quality predictions that can support policy decisions, public health measures, and early warning systems for urban populations.

4.2. Data pre-processing

Data pre-processing is critical in ensuring the accuracy, consistency, and reliability of the dataset used for air quality modeling. Given the complexity of air pollution data, which consists of multiple meteorological and pollutant variables, thorough data cleaning, transformation, and feature engineering were performed before applying machine learning models [40]. Since real-world data often contains inconsistencies, missing values, and redundancies, the dataset underwent a comprehensive cleaning and transformation process before being used for predictive modeling.

The first step in pre-processing involved removing unnecessary features that did not contribute significantly to the prediction model or contained excessive noise [41]. Several attributes such as City, Relative Humidity (RH), MP-Xylene, Solar Radiation (SR), Barometric Pressure (BP), Vertical Wind Speed (VWS), and Xylene were identified as non-essential and subsequently dropped from the dataset. The rationale behind this step was to enhance computational efficiency and improve model interpretability by eliminating redundant and irrelevant information. Once feature selection was completed, an initial exploratory data analysis (EDA) was conducted to detect duplicated records, missing values, and other inconsistencies. It was observed that while the dataset did not contain duplicate entries, certain columns had missing values. Missing data is a common issue in environmental datasets due to sensor malfunctions, data transmission errors, or temporary station outages. To address this, the missing values were imputed using mean substitution, where the average of the respective column replaced missing values. This method preserves the overall statistical properties of the dataset and ensures that no artificial trends are introduced.

After handling missing values, the next critical step was outlier detection and removal to ensure data integrity. Since extreme values can disproportionately affect machine learning models, a *Z*-score based outlier removal technique was applied. This method involves calculating the standard deviation of each feature and removing data points that deviate significantly from the mean (typically beyond three standard deviations). In particular, PM10 levels exhibited high variability, making it essential to filter out extreme values that could distort predictive performance. This step ensured that the dataset retained meaningful variations in air quality levels without being affected by occasional spikes caused by external factors such as sudden industrial emissions or extreme weather conditions.

To further improve model accuracy, a multicollinearity analysis was performed using the Variance Inflation Factor (VIF) to detect highly correlated independent variables. Multicollinearity occurs when predictor variables exhibit strong correlations with each other, leading to instability in regression models and difficulties in interpreting feature importance. High VIF values indicate redundancy among variables, which can negatively impact the reliability of predictions. Based on the VIF analysis, attributes such as Wind Speed (WS), Wind Direction (WD), Rainfall (RF), and Total Rainfall (TOT-RF) were identified as highly correlated with other meteorological factors and were consequently removed. Eliminating multicollinear features ensures that the remaining predictors contribute uniquely to the model, improving both interpretability and performance.

Another crucial transformation was applied to PM10 levels, which exhibited a right-skewed distribution, meaning extreme values were more common on the higher end of the scale [42,43]. To normalize this distribution and reduce skewness, a logarithmic transformation was applied. Log transformation is a widely used technique in time-series forecasting and regression analysis, as it helps stabilize variance and make data more normally distributed. By applying this transformation, the predictive models became more robust to variations in PM10 levels, thereby improving overall accuracy.

Following data cleaning and transformation, feature scaling was applied to ensure uniformity across all predictor variables. Since different attributes in the dataset had varying scales (e.g., temperature measured in degrees Celsius, pollutant concentrations in micrograms per cubic meter), Min-Max normalization was used to standardize all features to a common range. Feature scaling is particularly important for machine learning algorithms that rely on gradient-based optimization, such as decision trees, random forests, and gradient boosting models. Without proper scaling, variables with larger numerical ranges could dominate the learning process, leading to biased predictions.

Once the dataset was cleaned, transformed, and scaled, it was prepared for model training by splitting into predictor variables (X) and the target variable (y). The target variable in this case was log-transformed PM10 values, while the predictor variables included a subset of air pollutants and meteorological parameters, namely PM2.5, NO, NO_2, NOx, SO_2, CO, Ozone, Temperature, Benzene, and Toluene. The dataset was optimized for efficient and accurate air quality forecasting by retaining only the most relevant features. The final dataset was then ready for machine learning model development, ensuring optimal performance while minimizing noise and redundancy.

5. Reason behind selecting these models for this research

Here, we used three distinct supervised machine learning models to forecast the PMs concentrations in various regions of Maharashtra, including Aurangabad, Mumbai, Nashik, Pune, and Thane, among others. We've employed supervised techniques such as Random Forest, Decision Tree, XG Boos. To increase the precision of these models, we've used Grid Search CV to select the best features.

The selection of machine learning models—Decision Tree, Random Forest, and XGBoost—was guided by their relevance, interpretability, and widespread application in regression tasks like air quality prediction. Each of these models offers distinct advantages for this study:

1. **Decision Tree**: A simple yet effective model for regression tasks, chosen for its interpretability and ability to capture non-linear relationships in the data.
2. **Random Forest**: An ensemble learning method that improves upon Decision Trees by reducing overfitting and providing robust performance through bagging. Its capability to handle high-dimensional data and missing values makes it highly applicable for air quality prediction.
3. **XGBoost**: A state-of-the-art gradient boosting algorithm that is well-suited for tabular datasets with complex patterns. Its efficiency and ability to handle interactions between features made it an ideal choice for capturing intricate relationships in air quality data.

The inclusion of these models ensures a diverse evaluation of classical machine learning techniques, with varying levels of complexity and performance, providing a comprehensive baseline for comparison against the deep learning model, LSTM.

LSTM was included for its ability to capture temporal dependencies and long-term patterns inherent in time-series data, such as PM2.5 and PM10 measurements [44]. This comparative analysis highlights the strengths and limitations of traditional machine learning versus a deep learning approach, offering valuable insights into their applicability for air quality prediction tasks.

6. Implementation details

Following the data pre-processing phase, the next step in the research involved predicting air quality using supervised Machine Learning models. This phase was divided into two sub-steps: first, predicting air pollution levels using traditional ensemble learning algorithms such as Decision Tree, Random Forest, and Gradient Boosting (XGBoost), and second, using deep learning techniques like LSTM to capture temporal dependencies in the data. This section focuses on the first phase, where we employed three ensemble-based machine learning models to build predictive models for air pollution forecasting.

The primary goal of this stage was to leverage machine learning to predict PM2.5 concentrations based on historical pollution data and meteorological conditions. The dataset was divided into training (70 %) and testing (30 %) subsets using train-test splitting, ensuring that the models learned from past air quality trends and were evaluated on unseen data. The feature set (X) consisted of various air quality parameters and meteorological factors, while the target variable (y) represented PM2.5 concentrations, which is a crucial determinant of air quality.

6.1. Decision Tree Regressor

The first machine learning model used for prediction was the Decision Tree Regressor, which operates by recursively splitting the dataset based on feature importance. The Decision Tree was initialized with controlled hyperparameters, including a maximum depth of 5, a minimum sample split of 3, and a minimum leaf size of 2 to prevent overfitting. The model was trained on the training dataset (xtrain, ytrain) and evaluated using Root Mean Squared Error (RMSE), Mean Squared Error (MSE), and R^2 score. The decision tree model performed significantly better in capturing some non-linearity in the dataset. However, it was prone to overfitting, as Decision Trees tend to memorize the training data rather than generalize patterns effectively.

To improve the performance of the Decision Tree model, hyperparameter tuning was performed using GridSearchCV, a cross-validation-based optimization technique. The grid search tested multiple configurations of tree depth, number of features, and minimum sample split size to identify the best-performing model. After fine-tuning, the optimized Decision Tree model showed increased accuracy on test data, reducing the generalization error and improving predictive stability.

6.2. Random Forest Regressor

While Decision Trees offer improved accuracy, they suffer from high variance, meaning that small changes in data can lead to drastically different results. To mitigate this, a Random Forest Regressor was implemented. Unlike a single Decision Tree, a Random Forest builds multiple trees and aggregates their predictions to enhance stability and reduce overfitting. The model was trained using 100 trees (n_estimators = 100), a maximum depth of 7, and a minimum split size of 7, ensuring a balanced model that neither underfits nor overfits the data.

After training, the Random Forest model demonstrated superior performance over the Decision Tree model. Since it leverages feature bagging and random selection of subsets, it could generalize better to unseen data, achieving a higher R^2 score and lower RMSE. However, to further optimize the model's performance, RandomizedSearchCV was employed to fine-tune hyperparameters. Unlike Grid Search, Randomized Search explores a wider range of hyperparameter values, making it computationally more efficient while still identifying optimal model configurations. The optimized Random Forest model achieved even higher predictive accuracy, confirming the effectiveness of ensemble learning in air quality forecasting.

6.3. Gradient Boosting (XGBoost) Regressor

The final machine learning model employed was Gradient Boosting (XGBoost). This powerful tree-based boosting algorithm improves upon traditional Decision Trees and Random Forests by sequentially correcting errors in previous iterations. Unlike Random Forest, which builds trees independently, XGBoost builds trees iteratively, adjusting for residual errors in each iteration to refine predictions. The Gradient Boosting model was initialized with 100 trees (n_estimators = 100), a learning rate of 0.1, and a maximum depth of 7, ensuring a balance between learning complexity and generalization.

After training, XGBoost demonstrated the highest predictive performance among all models, outperforming both Decision Trees and

Random Forest. Since Gradient Boosting prioritizes misclassified data points, it was able to detect subtle variations in pollutant levels, leading to a more refined predictive model. To maximize accuracy, hyper-parameter tuning was performed using RandomizedSearchCV, optimizing parameters such as the learning rate and tree depth. The fine-tuned XGBoost model achieved the lowest RMSE and highest R^2 score, solidifying its effectiveness in air quality prediction.

6.4. Comparison of ML models

The performances of the three machine learning models were evaluated using standard regression metrics, including Root Mean Squared Error (RMSE), Mean Squared Error (MSE), and R^2 score. The results highlighted a clear progression in predictive accuracy:

- Decision Tree Regressor: Captured basic patterns but was prone to overfitting.
- Random Forest Regressor: Improved generalization and reduced variance by averaging multiple trees.
- XGBoost: Demonstrated the best performance by sequentially refining predictions and minimizing residual errors.

To further validate the effectiveness of the trained models, predicted values were compared with actual pollutant levels, and the results were exported as an Excel file (decision_tree.xlsx) for additional analysis. The high accuracy of the XGBoost model confirmed the potential of machine learning in air quality forecasting, providing a data-driven approach for monitoring and mitigating air pollution.

7. Transition to Deep Learning-Based Prediction (LSTM)

While Decision Trees, Random Forest, and XGBoost provided strong predictive capabilities, they were limited in their ability to capture temporal dependencies within the dataset [45]. Since air pollution levels fluctuate over time due to seasonal changes, human activities, and weather conditions, a time-series-based deep learning approach is required to model these patterns more effectively.

The next phase of this research explores Long Short-Term Memory (LSTM) networks, a specialized form of Recurrent Neural Networks (RNNs) designed for sequential data processing [46]. Unlike traditional ML models that rely solely on historical pollutant levels as independent variables, LSTM models utilize memory gates to retain long-term dependencies, making them highly suitable for time-series forecasting. The following section delves into the implementation of LSTM networks for air quality prediction, comparing their performance against the traditional machine learning models discussed above [47].

A deep sequential neural network was designed to build an LSTM-based air quality prediction model using multiple LSTM layers, dropout layers, and dense layers to ensure a robust learning process. The model architecture was carefully structured to enhance its ability to capture short-term fluctuations and long-term trends in air quality data.

The initial layer of the LSTM model consists of an LSTM layer with 50 units, which learns patterns in sequential data while preserving temporal dependencies. To prevent overfitting, a Dropout layer with a dropout rate of 20 % was added. Dropout regularization helps reduce model overfitting by randomly deactivating neurons during training, thus improving generalization to unseen data. The model then includes additional LSTM layers with increasing neuron sizes to capture hierarchical time-dependent patterns in the air pollution dataset. Following the LSTM layers, a fully connected Dense layer with a single neuron was added to produce the final prediction of pollutant concentration levels. The model was compiled using the Adam optimizer, an adaptive gradient-based optimization algorithm that helps achieve efficient convergence. Mean Squared Error (MSE) was used as the loss function to minimize prediction errors.

The final LSTM model was structured as follows:

- **Input Layer**: Accepts time-series data in a reshaped format to conform to LSTM input requirements.
- **LSTM Layer 1**: Contains **50 LSTM units**, allowing the model to learn sequential dependencies.
- **Dropout Layer 1**: Regularizes the model by randomly disabling **20 % of neurons** to prevent overfitting.
- **LSTM Layer 2**: Contains **100 LSTM units**, adding depth to capture more intricate time-series relationships.
- **Dropout Layer 2**: A **20 % dropout rate** is applied again to enhance model generalization.
- **LSTM Layer 3**: Contains **150 LSTM units**, ensuring that long-range dependencies in air pollution data are effectively captured.
- **Dense Output Layer**: A **single neuron layer** that provides the final predicted PM2.5 concentration.

The model was trained on the **training dataset (X_train, y_train)** using **100 epochs** and a **batch size of 1**, ensuring that the model had sufficient opportunities to learn complex patterns in the data. Training was performed in a **batch-wise manner**, where each data sample was passed through the network individually to optimize the learning process.

8. Model evaluation and performance metrics

After training, the LSTM model was evaluated on the **test dataset (X_test, y_test)** to assess its predictive accuracy. The **model's performance** was measured using key statistical metrics, including:

- **Root Mean Squared Error (RMSE)**: Measures the **average error** in predictions by calculating the square root of the mean squared differences between actual and predicted values. Lower RMSE values indicate better predictive accuracy.
- **R^2 Score (Coefficient of Determination)**: Evaluates how well the predicted values match the actual values. An **R^2 score closer to 1** indicates a **strong correlation between predicted and observed pollutant levels**.

The trained model produced two sets of predictions: one for the training dataset (train_predict) and one for the test dataset (test_predict). To facilitate interpretability, inverse transformations were applied to both sets of predictions, converting them back to their original scale using the scaler's inverse transformation. This step was crucial as the input features had been normalized during pre-processing, and converting the predictions back to their original scale allowed for meaningful comparisons with actual PM2.5 concentrations.

9. Result and discussion

Predicting PM2.5 and PM10 concentrations is a difficult job that demands the application of advanced ML and DL techniques. Time-series analysis, regression models, and Long Short-term Memory (LSTM) are common approaches. As in the dataset, there are many outliers; therefore, first of all, we had to remove all the outliers, pre-process the dataset, and apply three machine learning Models, and we have taken the R2 score and RMSE as evaluation matrices. In all three regression models, we get good accuracy (R2 score) but also more error which is not applicable to predicting and forecasting PM2.5 and PM10. Hence, we applied the LSTM (long-short-term memory) model, which gave us better accuracy than ML models and also very little RMSE (error).

Results and Visualization of these models are as follows:

The below tables (Tables 1 and 2) represent the overall model accuracy (R2 score) and RMSE score for PM2.5 concentration. From Table 1., we can easily understand that LSTM model has the highest R2 score (accuracy) and lowest RMSE (error rate) among the used algorithms. Among the five cities, Nasik and Pune have the ideal values.

Table 1
Overall Model Accuracy (R2-Score).

Cities	Decision-Tree	Random-Forest	XG-Boost	LSTM
Aurangabad	0.6339	0.7163	0.7590	0.9976
Mumbai	0.7814	0.8573	0.8849	0.9970
Nasik	0.7685	0.8207	0.8374	0.9999
Pune	0.7486	0.8675	0.9104	0.9999
Thane	0.3488	0.5228	0.4952	0.9980

Table 2
Root Mean Square Error (RMSE).

Cities	Decision-Tree	Random-Forest	XG-Boost	LSTM
Aurangabad	9.1523	8.0573	7.4263	0.70
Mumbai	14.5923	11.7890	10.5887	0.54
Nasik	9.4004	8.2733	7.8788	0.15
Pune	13.4217	9.7438	8.0130	0.15
Thane	6.6213	4.6557	4.7886	0.20

- **Results for PM2.5:**

o Table 3 and Table 4 represents the values of R2 score and RMSE for PM10 concentration. Equivalently, LSTM has the highest model accuracy of 99 % and lowest error rate of 0.3 among used algorithms. Pune and Nasik are dominating with ideal numbers.

- **Results for PM10:**

- **Graphs of LSTM model:**

These all following visualizations are graph of lstm models applied to the dataset of different cities.

9.1. PM2.5 predictions in different cities

Among the below five graphs, Thane has a constant value of PM2.5 within 10–1100 and 1175–1400 data samples across time. While most of the cities are having fluctuation among value as increasing data samples across time. Figuratively, actual and predicted values are overlapping to each other in every graph. Fig. 1 demonstrates that there is a very minute (almost negligible) error rate among prediction of PM2.5 values across different cities.

9.2. PM10 predictions in different cities

As we move from left to right across the above graph as shown in Fig. 2, it illustrates that Aurangabad values, based on their data, represent lower PM10 values than those of the other four cities. Meanwhile, the PM10 readings in the four cities fluctuate as we walk through the samples. The PM10-value in the Aurangabad figure, which contains about 1000 data samples, is the highest of the five cities.

10. Challenges

One of the main causes of air pollution, which has grown to be a major environmental concern worldwide, is particulate matter (PM) [48]. Two of the most harmful air pollutants that can result in respiratory and cardiovascular issues are PM 2.5 and PM 10. As a result, forecasting PM 2.5 and PM 10 concentrations has become a crucial field of research in environmental science. The difficulties and future prospects of forecasting PM 2.5 and PM 10 are examined in this article. It is challenging to forecast PM2.5 and PM10 concentrations with LSTM models due to the complexity and nonlinearity of the air pollution data. The advancement of machine learning techniques has allowed researchers to forecast significant increases in these concentrations with LSTM models.

Table 3
Overall Model Accuracy(R2-Score).

	Linear-Regression	Decision-Tree	Random-Forest	XG-Boost	LSTM
Aurangabad	0.5221	0.5451	0.7689	0.7700	0.9980
Mumbai	0.7469	0.7549	0.8437	0.8735	0.9977
Nasik	0.1807	0.7014	0.7625	0.7745	0.9940
Pune	0.5608	0.8794	0.8556	0.9552	0.9938
Thane	0.3535	0.5155	0.6587	0.7425	0.9965

Table 4
Root Mean Square Error (RMSE).

	Linear-Regression	Decision-Tree	Random-Forest	XG-Boost	LSTM
Aurangabad	0.0677	18.6580	13.2988	13.2681	0.37
Mumbai	0.0916	24.7999	19.8058	17.8134	0.73
Nasik	0.3511	16.7482	14.9374	14.5558	0.74
Pune	0.1919	13.3004	14.5504	8.1049	2.61
Thane	0.4167	52.8907	44.3926	38.5599	4.49

The complexity of air pollution Forecasting PM 2.5 and PM 10 amount may be difficult since air pollution is a complicated process. Air pollution is caused by several things, including transportation, weather, industrial pollutants, and natural phenomena like wildfires and dust storms [49].

Inaccurate Data: The precision of the data utilised significantly impacts how well PM 2.5 and PM 10 levels are predicted. But a major barrier still exists regarding the accessibility of reliable data, especially in underdeveloped countries. Due to inadequate monitoring systems, a lack of resources, and poor data sharing among companies, it is challenging to get reliable statistics.

Despite significant research on the impacts of air pollution on human health, it is still unclear how PM 2.5 and PM 10 are produced and dispersed. It is challenging to estimate the amounts of these contaminants with any degree of accuracy without a good understanding of these mechanisms [50].

Source Variability: Because the sources of PM 2.5 and PM 10 are so unpredictable and prone to sudden fluctuation, it is challenging to estimate the levels. For instance, increased traffic can cause a sudden rise in PM 2.5 and PM 10 levels, and forest fires can have a substantial impact on the quality of the air in the area.

One of the biggest challenges in utilizing LSTM models to forecast PM2.5 and PM10 concentrations is a lack of data. Air pollution data is not always easy to find, and even when it is, it might not cover enough ground or be insufficient [51].

The accuracy of data on air pollution can also be a concern because it can be affected by a number of things, including sensor malfunction, measurement blunders, and calibration issues.

11. Future scope

In this study, we compared traditional machine learning algorithms (Decision Tree, Random Forest, XGBoost) with a deep learning model (LSTM) for predicting air quality indices (AQI) based on PM2.5 and PM10 levels. While the findings demonstrate the superiority of LSTM in terms of R2R^2R2 and RMSE, there are several avenues for further research and development to enhance and generalize the study:

Incorporation of Additional Deep Learning Architectures

- Future studies can explore other advanced deep learning models like **GRU (Gated Recurrent Unit)**, **Transformer-based models**, or **CNN-LSTM hybrids** to analyze if they can outperform LSTM for AQI prediction.
- GRUs are computationally simpler and may provide similar or better performance with reduced training time.

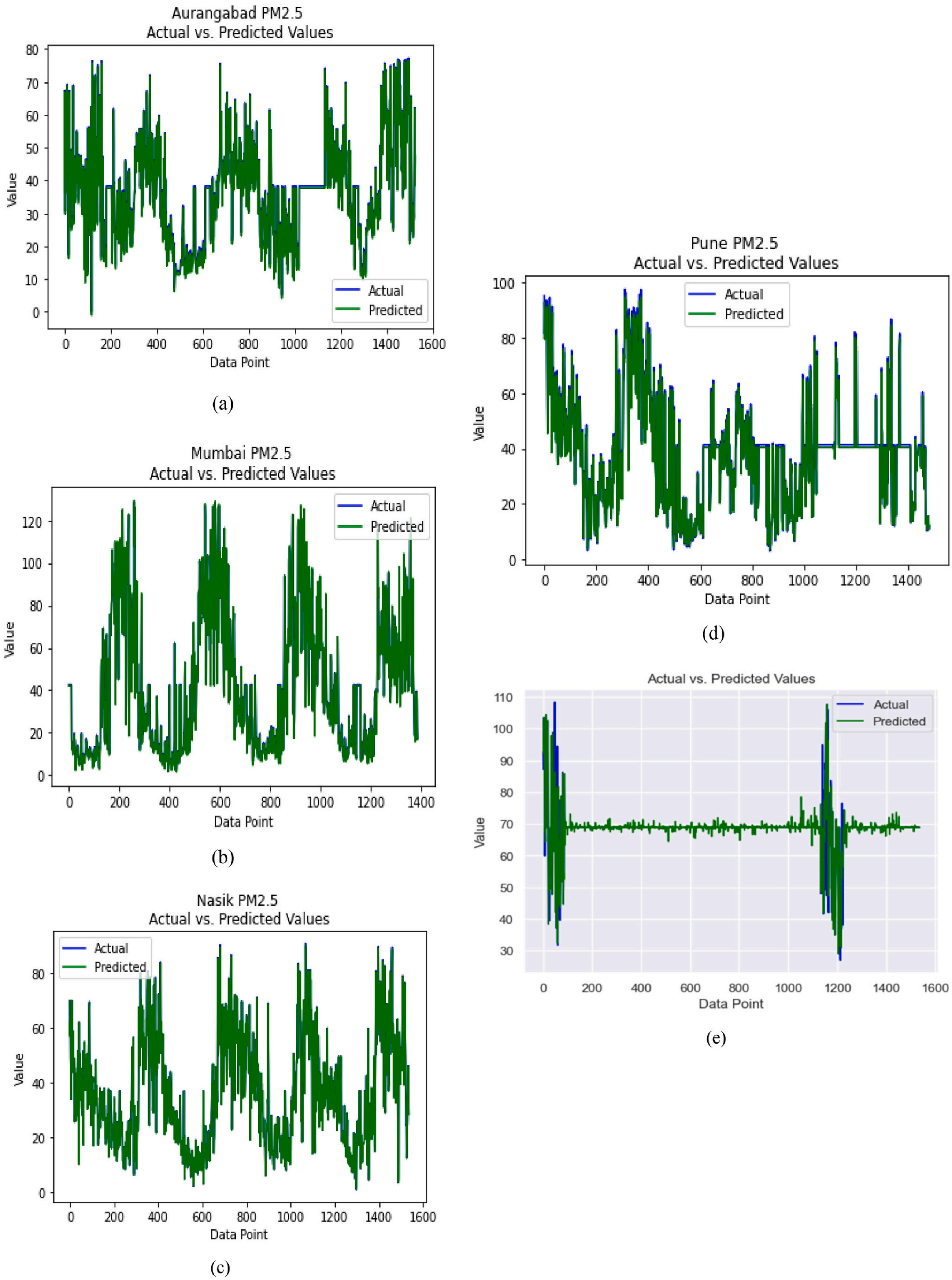

Fig. 1. Plots between value of PM2.5 and range of data points in (a) Aurangabad, (b) Mumbai, (c) Nasik, (d) Pune and (e) Thane.

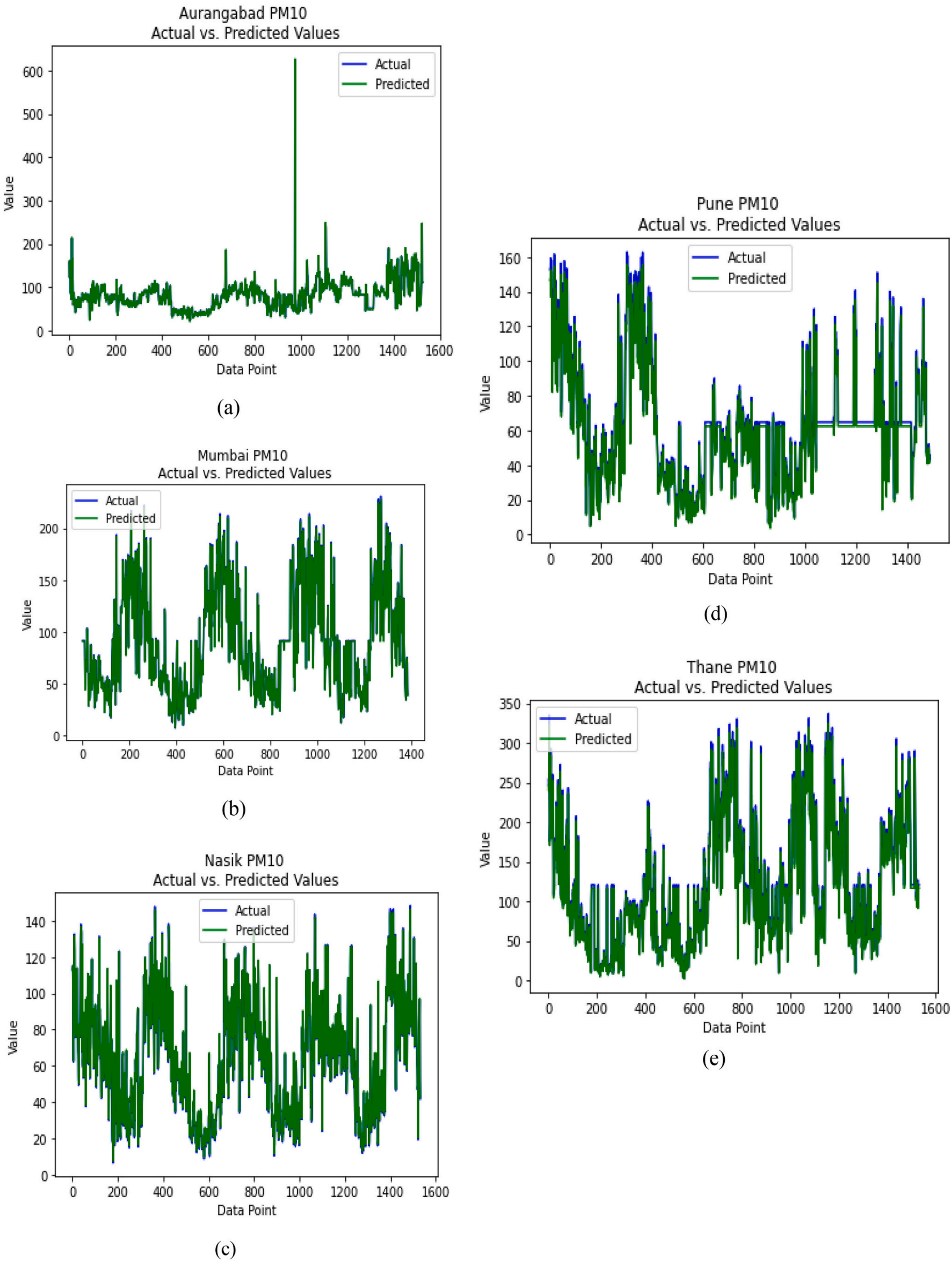

Fig. 2. Plot between value of PM10 and range of data points across Aurangabad (a), Mumbai (b), Nasik (c), Pune (d), and Thane (e).

- Transformer-based models (like those used in natural language processing) have recently shown promise in handling sequential data and can be adapted for time-series forecasting.

Multimodal Data Integration

- This research focused on PM2.5 and PM10 as key input features. However, incorporating additional environmental factors like temperature, humidity, wind speed, and atmospheric pressure could provide richer insights and improve prediction accuracy.
- Multimodal datasets combining sensor data, satellite imagery, and geographic information could further enhance model performance.

Geographic and Temporal Generalization

- The current study is limited to the dataset from a specific region (e.g., Taiwan EPA data). Expanding the analysis to multiple regions with diverse environmental conditions and seasonal variations could validate the robustness and generalizability of the models.
- Temporal generalization could involve testing the model on more extended forecasting horizons (e.g., predicting AQI days or weeks ahead) or transferring the model to other temporal datasets.

Explainability of Models

- For deep learning models like LSTM, explainability is often a challenge. Future work can leverage **explainable AI (XAI)** techniques to interpret how the model makes predictions and identify key contributing factors.
- Methods such as **SHAP (SHapley Additive exPlanations)** or **LIME (Local Interpretable Model-agnostic Explanations)** could be applied to understand the model's decision-making process and improve trustworthiness in practical applications.

Real-Time Deployment

- While this study focused on offline predictions, deploying the LSTM model in a **real-time system** for continuous monitoring and prediction of AQI could significantly impact air quality management.
- This would involve optimizing the model for faster inference and ensuring it can handle streaming data efficiently.

Incorporating Domain Knowledge

- Incorporating domain knowledge, such as the impact of pollution sources (e.g., industrial emissions, traffic patterns) or regulatory policies, could enhance the accuracy and practicality of the models.
- Integrating these domain-specific insights into the feature engineering process or directly into the model could make the predictions more actionable.

Uncertainty Quantification

- Air quality prediction inherently involves uncertainties due to the dynamic nature of environmental factors. Future research could focus on quantifying the uncertainty in predictions using probabilistic methods or Bayesian frameworks for both ML and DL models.
- This would help policymakers and environmental agencies make more informed decisions.

Comparison with Ensemble and Hybrid Models

- Hybrid models combining the strengths of ML and DL techniques could be explored. For example:
  - Use XGBoost or Random Forest for feature selection and preprocessing, followed by LSTM for prediction.
  - Combine CNNs with LSTMs to extract spatial and temporal features from datasets like satellite imagery.

Inclusion of Socioeconomic and Health Data

- Expanding the dataset to include **socioeconomic factors** (e.g., population density, industrial zones) and **health-related data** (e.g., hospital admissions due to respiratory illnesses) could provide a more comprehensive framework for understanding the impact of air quality predictions.
- Such models could serve as early warning systems to minimize health risks.

Model Optimization and Energy Efficiency

- Deep learning models like LSTM are computationally intensive. Future research could explore optimizing the training and inference processes to reduce computational costs and energy consumption.
- Techniques like **model pruning**, **quantization**, and **knowledge distillation** could make these models more resource-efficient.

Policy Impact and Real-World Applications

- The insights gained from predictive modeling can be applied to **policy-making**, such as identifying critical pollution hotspots and prioritizing regulatory actions.
- Collaborations with governmental agencies and environmental organizations could lead to deploying these models in decision-making systems.

Open-Source Tools and Community Collaboration

- Publishing the dataset, code, and trained models as part of an **open-source initiative** would enable other researchers to reproduce and extend the study.
- Community collaboration can lead to improvements in the methodology and encourage benchmarking against other models.

12. Conclusion

In this study, we employ regression models such as decision tree, random forest, and XG boost in addition to the deep learning model LSTM. The results of the study show that LSTM models can successfully forecast PM2.5 and PM10 concentrations. Our work demonstrated that LSTM models can accurately and successfully anticipate PM2.5 and PM10 concentrations. R2-Square and RMSE, among other measures, were used to gauge how well the LSTM model worked. Our findings demonstrated that when it came to estimating PM2.5 and PM10 concentrations, the LSTM model outperformed traditional statistical methods. The capability of the LSTM to capture the relationships between the input data is one of its main advantages, and it is particularly helpful for forecasting air pollution concentrations.

In addition, LSTM models can accommodate non-linear relationships between the input and output variables, which are frequently present in air pollution data. Our study shows that LSTM models can properly forecast PM2.5 and PM10 concentrations, offering a useful tool for managing air quality and formulating policies. The outcome of this examination can be applied to developing methods that effectively reduce air. But it's crucial to remember that our study has some limitations. The model's effectiveness may change depending on the setting, timeframe, and data used. Additional investigation is necessary to assess the LSTM model's generalizability in predicting air pollution concentrations in different places and to pinpoint the critical variables that affect the model's performance.

The primary focus of this research was to compare classical machine learning models, such as Decision Trees, Random Forest, and XGBoost, with a deep learning model to evaluate their effectiveness in predicting PM2.5 and PM10 air quality indices. LSTM was specifically chosen because it is particularly well-suited for time-series data, which is a key characteristic of air quality measurements. Its ability to capture temporal dependencies and manage long-term correlations in sequential data makes it a robust choice for this application.

While other deep learning models, such as CNNs or GRUs, could potentially be explored, LSTM has been extensively validated in similar environmental prediction studies and is recognized for its high performance in handling time-series data with long dependencies. Given the computational complexity and resource constraints associated with training multiple deep learning models, the focus was to use LSTM as a benchmark to assess whether deep learning could outperform traditional ML models for this specific task.

The superior results of LSTM in terms of R^2 and RMSE further confirm its effectiveness for this problem. However, as a future direction, we acknowledge that incorporating other deep learning architectures, such as GRUs or hybrid models, could provide additional insights and enrich the comparative analysis.

CRediT authorship contribution statement

Pranshu Patel: Writing – original draft. **Swara Patel:** Writing – review & editing. **Kanish Shah:** Writing – review & editing, Validation. **Kedar Desai:** Data curation, Conceptualization. **Samrat Patel:** Supervision, Methodology. **Manan Shah:** Writing – review & editing. **Samir Patel:** Writing – review & editing.

Consent for publication

Not applicable.

Ethics approval and consent to participate

Not applicable.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.enceco.2025.07.001.

Data availability

All relevant data and material are presented in the main paper.

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